IFM2B10: Group Project Proposal

Quantum Coders

Team Members

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Project Idea: OPALÉRA

The house of every gem

Name of the system

OPALÉRA

E-Business category

Business-to-Consumer (B2C).

Business case description

Jewellery is part of beauty in this new generation, but buying it can be risky: shoppers can encounter scams, such as being sold fake diamonds, and jewellery is often very expensive for the average person on a basic salary.

OPALÉRA is an online store that will solve these problems by selling jewellery accessories — mainly necklaces, earrings and pendants — at affordable prices. The platform will be available to anyone buying jewellery for themselves or their loved ones. It will offer a wide range of items to pick from, and it will be easy to navigate and to find information and details about each item.

Possible users

- **Customers (clients) —** browse the catalogue and buy jewellery for themselves or their loved ones.
- **Jewellery manufacturers —** use the platform's sales data to analyse the most-bought items so they can manufacture more of the favourite products.
- **Administrator —** manages product listings, verifies that items are genuine, and maintains the platform.

Value added by the project

- Cheaper prices compared to other jewellery sellers, making jewellery accessible to people on a basic salary.
- A safe, trusted place to buy jewellery, reducing the risk of scams such as fake diamonds.
- A wide range of items with clear information and details, on a platform that is easy to navigate.
- Demand insights for manufacturers, who can see which items are bought most.

What sets it apart from current systems — and why we chose OPALÉRA

We chose OPALÉRA because it attacks a real, high-stakes consumer problem — counterfeit stones and boutique mark-ups — and because no current jewellery seller combines the following five features on one platform:

- **Trust engineered into every piece.** A rating and review system where each review is tied to an actual purchase and badged "Verified purchase", so buyers see the genuine experience of others and the true quality of each item before they pay. Existing jewellery sites offer either no reviews or unverifiable ones.
- **A certificate of authenticity on every product page.** Each piece carries its own certificate — stone, metal and hallmark verified by the administrator before listing — backed by a full money-back promise. This answers the fake-diamond scam problem in

our business case directly, on the page where the buying decision happens.

- **A customisation studio with craftsmanship rules.** Customers can change a piece's design pattern, add a hand-engraved inscription (up to 30 characters) and add a pendant to any necklace — while the stone itself can never be altered and the maison's most precious pieces accept inscriptions only. Rule-based personalisation of this kind is absent from mainstream jewellery retailers.
- **Transparent affordability.** Every piece displays the comparable boutique price alongside ours and an interest-free instalment breakdown, so the promise of "jewellery on a basic salary" is visible on every product rather than only in marketing copy.
- **Data that works both ways.** A live analytics view of the catalogue — pieces per collection, price distribution and most-loved items — gives jewellery manufacturers the demand insights identified in our user analysis, while helping shoppers navigate the range.

These differentiators are not aspirational: a working front-end prototype of the store — ratings and verified reviews, certificates, the customisation studio, wishlist, shopping bag and customer accounts — has already been built on the project dataset, demonstrating that the concept is deliverable within the module timeline.

Technologies the team intends to use

- HTML5 — website creation.
- C#, Java and/or Visual Basic — application development.
- Python — database development and management.

Dataset for this project

https://www.kaggle.com/datasets/harshjangid0015/jewelry-database